RAG SYSTEM ARCHITECTURE

KNOWLEDGE INDEXING & RETRIEVAL TECHNICAL MANUAL

DOCUMENT ID	TARGET ENGINE	CHUNK STRATEGY	CLASSIFICATION
RAG-SPEC-2026	Vector Pipeline	Semantic Hybrid	Technical Baseline

1. ENTERPRISE RAG LIFECYCLE OVERVIEW

Retrieval-Augmented Generation (RAG) bridges non-parametric LLM capabilities with dynamic external knowledge bases. The ingestion phase parses unstructured assets into normalized, vector-embedded semantic blocks. Query time execution performs hybrid dense/sparse retrieval, reranking, and dynamic prompt injection for factual groundedness.



2. INGESTION & PREPROCESSING MECHANICS

Document Cleansing & Normalization

Input PDFs and markdowns undergo structural stripping to clean boilerplate headers, footers, and redundant linebreaks. Character sets are standardized to UTF-8 with standard whitespace normalization.

Hierarchical Recursive Chunking

Documents are split using dynamic boundaries: structural headings first, followed by paragraph blocks, targeting an optimal token window of 256–512 tokens with 15% overlap ratio.

Metadata Augmentation

Every chunk receives key metadata payloads including document UUID, section hierarchy, page numbers, created timestamp, and access control tags (ACLs) for filtered retrieval.

Embedding Generation

Text chunks pass through dense bi-encoder models producing normalized float32 vectors. High-dimensional outputs (e.g., 1536d) are formatted for fast cosine metric calculation.

3. CHUNKING & INDEXING BENCHMARK CONFIGURATIONS

Strategy	Window Size	Overlap	Primary Use Case	Retrieval Precision
Fixed Character	500 chars	50 chars	Unstructured raw text logs	Moderate (62%)
Recursive Character	512 tokens	64 tokens	General documentation & web pages	High (84%)
Semantic Boundary	Dynamic	Variable	Technical reports & chapters	Very High (91%)
Hierarchical Parent-Child	2048 / 256 tokens	32 tokens	Complex legal & compliance PDFs	Optimal (95%)
Agentic Dynamic	Context Driven	Adaptive	Multi-step research corpora	Optimal (96%)

4. ADVANCED HYBRID RETRIEVAL FRAMEWORK

To maximize search recall and precision, state-of-the-art implementations employ hybrid search combining dense semantic vectors with sparse lexical indices (BM25). The retrieved candidate sets are merged using Reciprocal Rank Fusion (RRF) before entering cross-encoder re-ranking.

> **Reciprocal Rank Fusion (RRF) Score Calculation**
> Given rank `r(d)` for document `d` across system metrics `M`:
> `RRF_Score(d) = ∑ (m ∈ M) [ 1 / ( k + r_m(d) ) ]` where constant `k ≈ 60`.

5. OPTIMIZATION AND EVALUATION PROTOCOL

Evaluating RAG pipelines requires non-reference metrics to continuously monitor search quality and hallucination rates in production contexts:

- **Faithfulness:** Verifies if the answer is completely grounded in the retrieved context chunks.
- **Answer Relevance:** Measures alignment between the user prompt and the generated response.
- **Context Precision:** Assesses whether top-ranked retrieved passages contain relevant answers.
- **Context Recall:** Checks if all ground-truth information required was successfully fetched.